

DataInsight Analytics

Predictive Model Explainability Report

*Partial Dependence Plot Analysis for Bike Rentals
& House Price Prediction*

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Executive Summary

This report has been prepared by DataInsight Analytics to provide our client with a comprehensive explainability analysis of the deployed predictive models. As your trusted analytics partner, our goal is not only to deliver accurate predictions, but also to ensure you fully understand the drivers behind each model's output - enabling confident, data-driven business decisions.

We have applied Partial Dependence Plots (PDPs), a model-agnostic explainability technique, to two predictive use cases:

- **Use Case 1** - Bike rental demand forecasting (1D and 2D PDP analysis)
- **Use Case 2** - Residential house price prediction (1D PDP analysis)

Both models are built on Random Forest Regressors, known for their high predictive performance. The PDP analyses presented here reveal how individual features - and their interactions - influence model predictions, while keeping all other variables constant.

Key findings are summarized below:

Bike Rental Model	House Price Model
• Temperature is the strongest driver of rental demand • Optimal conditions: 20-30°C, low humidity • High humidity (>70%) and strong winds reduce rentals • Demand has grown consistently since 2011	• Living area (sqft_living) is the top price predictor • Bathrooms show a strong positive effect on price • Bedroom count has a non-linear, nuanced relationship • More floors moderately increases predicted price

Contents

Executive Summary	2
1. Introduction	4
1.1. Model Classification Framework	4
1.2. Why Partial Dependence Plots?	5
2. One-Dimensional PDP: Bike Rental Demand.....	6
2.1. Objective	6
2.2. Methodology.....	6
2.3. Data Processing	6
2.4. Feature Importance	7
2.5. PDP Results and Interpretation	7
3. Two-Dimensional PDP: Temperature × Humidity	9
3.1. Objective	9
3.2. Methodology.....	9
3.3. Results and Interpretation.....	9
4. One-Dimensional PDP: House Price Prediction	11
4.1. Objective	11
4.2. Methodology.....	11
4.3. Feature Importance	11
4.4. PDP Results and Interpretation	12
5. Conclusions & Recommendations.....	14
5.1. Bike Rental Demand Model	14
5.2. House Price Prediction Model.....	14
5.3. On the Value of Explainable AI	15

1. Introduction

As part of our engagement with your organization, DataInsight Analytics has developed and deployed a series of predictive models designed to support data-driven decision-making across your business lines. Our commitment extends beyond model accuracy: we believe that for a model to generate true business value, its underlying logic must be transparent and interpretable by the people who rely on it.

To achieve this, we have applied Partial Dependence Plots (PDPs), a powerful and widely recognized model-agnostic explainability method. PDPs allow us to visualize and communicate the marginal effect of each input feature on the model's predictions, independently of all other features.

1.1. Model Classification Framework

To provide context for our approach, it is useful to understand how predictive models are broadly classified:

Category	Description & Examples
Interpretable Models	Models whose internal logic is directly understandable. Examples: Logistic Regression, Decision Trees, RuleFit.
Black Box Models	Highly complex models where internal logic is opaque. Examples: Neural Networks, Support Vector Machines (SVM), Gradient Boosting.
Model-Specific Methods	Explanation techniques that only apply to one model type. Example: regression coefficients (only valid for linear models).
Model-Agnostic Methods	Explanation techniques applicable to ANY model, regardless of complexity. Only the inputs and outputs are needed. Examples: PDP, SHAP, LIME.

1.2. Why Partial Dependence Plots?

For this engagement, we have selected PDPs as our primary explainability tool because they are model-agnostic - they can be applied to both interpretable and black-box models - and they produce clear, intuitive visualizations that stakeholders can understand without deep technical knowledge.

The analyses in this report cover:

- 1D PDP: Bike rental demand - marginal effect of days since 2011, temperature, humidity, and wind speed on predicted rentals from a Random Forest model.
- 2D PDP: Bike rental demand - joint interaction effect of temperature and humidity.
- 1D PDP: House price prediction - marginal effect of bedrooms, bathrooms, living area, sqft_lot, floors, and year built on predicted prices.

2. One-Dimensional PDP: Bike Rental Demand

2.1. Objective

This section presents the 1D Partial Dependence Plot analysis for the bike rental forecasting model. The goal is to understand and communicate to you, our client, how each individual feature affects predicted rental demand in isolation - while all other variables are held constant at their average values.

2.2. Methodology

Model Configuration
Dataset: Bike Sharing Dataset (day.csv) - daily rental records
Target Variable: cnt (total daily bike rentals)
Model: Random Forest Regressor (500 trees)
Model Fit: $R^2 = 88.48\%$ variance explained
Features Analyzed: days_since_2011, temperature (temp), humidity (hum), wind speed (windspeed)

2.3. Data Processing

Prior to modeling, the dataset was preprocessed as follows to ensure meaningful feature scales:

- Season encoded as binary dummy variables (season1, season2, season3).
- Weather conditions encoded: MISTY (weathersit = 2) and RAIN (weathersit = 3 or 4).
- Temperature denormalized to degrees Celsius (multiplied by 41).
- Humidity denormalized to percentage (multiplied by 100).
- Wind speed denormalized to km/h (multiplied by 67).
- Days since January 1, 2011 computed as a continuous time feature.

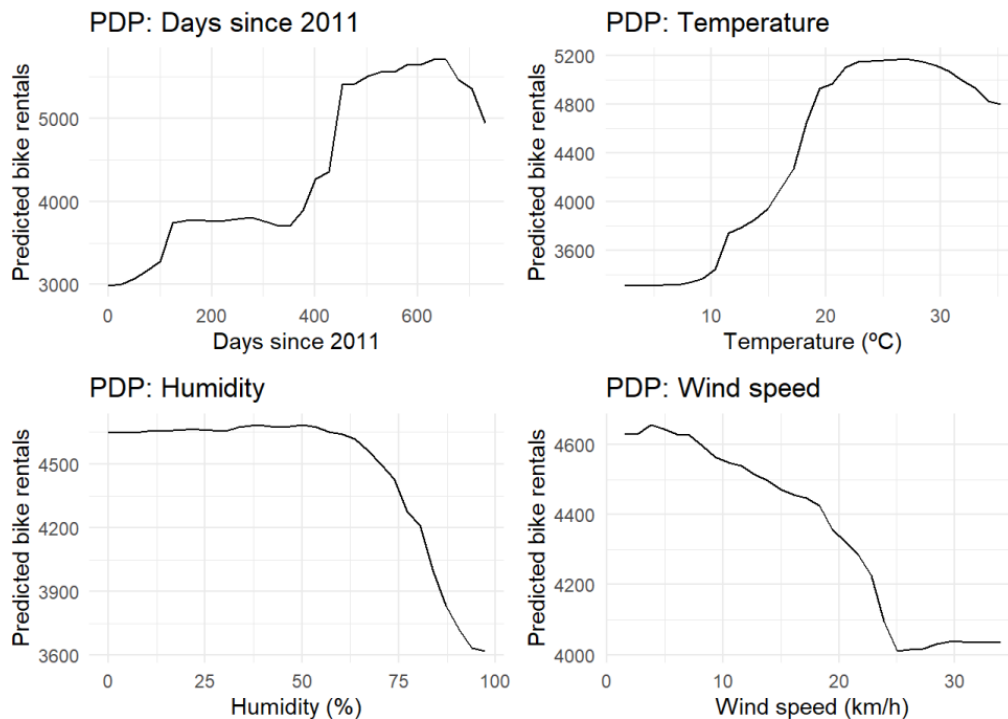
2.4. Feature Importance

The table below shows the importance of each feature in the trained Random Forest model, ranked by %IncMSE (percentage increase in Mean Squared Error when the feature is removed):

Feature	Importance (%IncMSE) - Higher = More Important
days_since_2011	77.23 - Most important feature; captures long-term demand growth
temp (Temperature)	44.80 - Second most important; strong effect on rentals
hum (Humidity)	32.89 - Significant negative driver at high levels
windspeed	19.40 - Moderate negative effect
MISTY	20.17 - Adverse weather reduces rentals
RAIN	19.35 - Strong negative effect on demand
season1	23.16 - Seasonal patterns captured

2.5. PDP Results and Interpretation

The following four PDPs were generated and are interpreted below. Each plot shows the average predicted rental count as a function of a single feature.



days_since_2011

The model captures a clear and consistent upward trend in rental demand over time, reflecting the organic growth of the bike-sharing service since its launch in 2011. This is followed by a slight stabilization or mild decline at later time periods, likely reflecting market saturation. From a business perspective, this confirms that the service has been successfully expanding and that long-term demand is strong.

Temperature

Temperature has a strong positive effect on rentals up to approximately 25°C, at which point demand slightly decreases. This non-linear relationship suggests that there is an optimal temperature range for bike rental activity. We recommend that your operations team take this into account for capacity planning: demand peaks during mild, warm days and drops off in extreme heat.

Humidity

Humidity exhibits a negative relationship with predicted rentals, with the effect becoming particularly pronounced above 70% relative humidity. This finding indicates that very damp conditions are a significant deterrent to cycling. This information can be used to anticipate lower-demand periods and adjust staffing or pricing strategies accordingly.

Wind Speed

Wind speed shows a consistently negative effect on bike rental predictions. As wind conditions increase, expected rentals decline, which is aligned with the physical discomfort of cycling in windy environments. While this variable is less impactful than temperature or humidity, it still contributes meaningfully to demand forecasting accuracy.

3. Two-Dimensional PDP: Temperature × Humidity

3.1. Objective

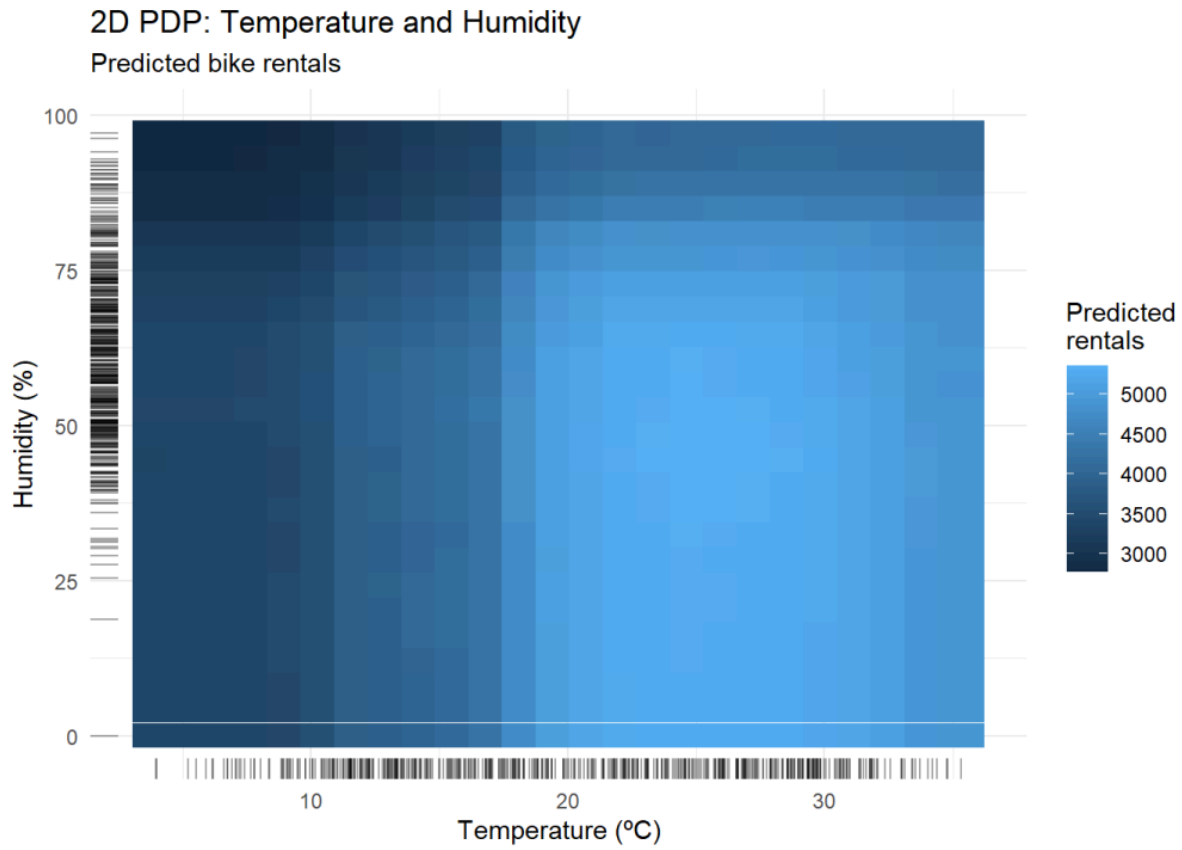
While the 1D PDPs in the previous section revealed the individual impact of temperature and humidity on bike rental demand, this section takes the analysis one step further by examining their joint interaction effect through a 2D Partial Dependence Plot.

3.2. Methodology

2D PDP Configuration
Variables analyzed: temp (Temperature, °C) and hum (Humidity, %)
A random subset of 500 observations was used to reduce computational cost
A 25×25 grid resolution was applied across the feature space
Visualization: Heatmap with rug plot showing actual data density

3.3. Results and Interpretation

The 2D heatmap reveals meaningful interaction patterns between temperature and humidity that cannot be identified from 1D analyses alone:



Condition	Predicted Demand	Business Implication
Low temp + high humidity	Lowest (~3,000)	Expect low demand; consider maintenance or rebalancing during these periods
High temp + low humidity	Highest (~5,000+)	Peak demand window; maximize fleet availability and staffing
High temp + high humidity	Moderate (~4,000-4,500)	Temperature's positive effect is partially offset by humidity; moderate demand expected

Overall, temperature is the dominant factor: rental demand increases sharply with warmth regardless of humidity. However, high humidity does partially dampen the positive effect of warm temperatures. This interaction insight allows your team to make more nuanced forecasts than either variable alone would provide.

4. One-Dimensional PDP: House Price Prediction

4.1. Objective

This section presents the explainability analysis for the residential house price prediction model. Using PDP methodology, we isolate and communicate the marginal effect of key structural property features on the model's predicted prices.

4.2. Methodology

Model Configuration
Dataset: King County House Sales (kc_house_data.csv)
Target Variable: price (residential sale price in USD)
Model: Random Forest Regressor (500 trees, trained on a 3,000-observation sample)
Model Fit: $R^2 = 56.54\%$ variance explained
Features Analyzed: bedrooms, bathrooms, sqft_living, sqft_lot, floors, yr_built

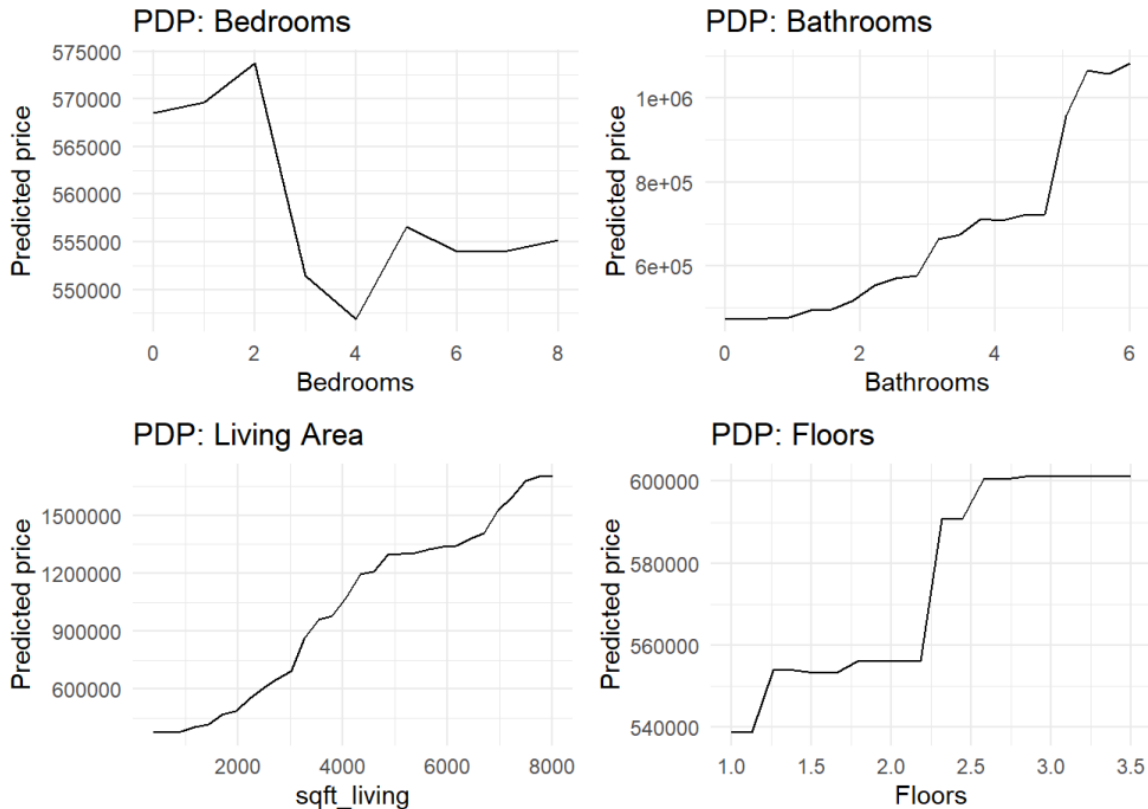
4.3. Feature Importance

The table below presents feature importance rankings from the trained model:

Feature	Importance (%IncMSE) - Higher = More Important
sqft_living (Living Area)	59.52 - Most influential structural feature by a wide margin
yr_built (Year Built)	42.82 - Strong influence; newer homes tend to command higher prices
bathrooms	23.22 - Significant positive price driver
sqft_lot (Lot Size)	19.88 - Moderate effect on price
floors	17.55 - Moderate contribution
bedrooms	12.82 - Least impactful; non-linear relationship observed

4.4. PDP Results and Interpretation

Below we present and interpret the PDP for each analyzed feature. These findings are directly actionable for your valuation and acquisition strategies.



Bedrooms

Contrary to intuition, the relationship between bedroom count and predicted price is non-linear. The model identifies homes with approximately 2 bedrooms as commanding the highest predicted prices, while additional bedrooms beyond that tend to stabilize or slightly reduce the predicted value. This may reflect the fact that adding bedrooms reduces other living spaces, or that very large-bedroom properties are often in lower-value segments. We recommend exercising caution when using bedroom count as a proxy for value.

Bathrooms

Bathrooms exhibit a clear and consistent positive effect on predicted house prices. Each additional bathroom is associated with meaningfully higher predicted values, with the effect accelerating at higher bathroom counts. This aligns with market expectations: bathrooms are a strong quality signal to buyers. For valuation purposes, bathroom count should be weighted heavily.

Living Area (sqft_living)

Living area is the single most influential structural feature in the model. Predicted prices increase steadily and substantially as living area grows, with the effect remaining consistent across the entire range of values. This finding strongly validates the use of square footage as a primary pricing driver in your models, and confirms that investment in larger properties is well-supported by the data.

Floors

The number of floors shows a moderate positive effect on predicted prices. The PDP curve displays a staircase pattern, reflecting the discrete nature of this variable (properties only have whole or half-floor increments in the data). Homes with more floors are generally predicted at higher prices, although the marginal gain decreases at higher floor counts. This is a useful input for new development pricing strategies.

5. Conclusions & Recommendations

The PDP analyses carried out by DataInsight Analytics have successfully illuminated the decision logic of both deployed predictive models. Below we summarize our key conclusions and the actionable recommendations we put forward for your consideration.

5.1. Bike Rental Demand Model

Key Finding	Recommendation
Temperature is the primary driver (optimal: 20-30°C)	Use weather forecasts to proactively scale fleet availability
Humidity above 70% significantly reduces demand	Integrate real-time humidity data into operational dashboards
High wind speed negatively impacts rentals	Consider wind advisories as a demand signal
Demand has grown steadily since 2011	Long-term growth trend supports continued fleet investment

5.2. House Price Prediction Model

Key Finding	Recommendation
sqft_living is the dominant price predictor	Prioritize living area in all valuation and acquisition workflows
Bathrooms have a strong, consistent positive effect	Weight bathroom count heavily in automated valuation models (AVMs)
Bedroom count shows a non-linear, counter-intuitive pattern	Avoid using bedroom count as a linear pricing proxy
More floors moderately increases predicted value	Account for floor count when pricing multi-story developments

5.3. On the Value of Explainable AI

The PDP technique has demonstrated its value as a bridge between model complexity and business understanding. By applying model-agnostic explainability methods, DataInsight Analytics is able to deliver the best of both worlds: the high predictive performance of Random Forest models, alongside the transparency needed for you to trust, validate, and act on those predictions.

We remain committed to supporting your team in leveraging these insights effectively. Should you require deeper analysis, additional features, or model updates, our team is available to discuss next steps at your convenience.

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May 10, 2026